

Axyl Carefoot-Schulz

Calgary, Alberta | [Linkedin](#) | [Github](#) | [Portfolio](#)

Professional Summary

Software Developer with experience in web development, system-level programming, and AI/ML projects. Built an internal wiki using React/Quartz to improve team resource sharing and developed an accessibility-focused resource portal for neurodivergent users. Experienced with C, Python, C#, Linux, automation, and building custom AI-driven applications. Eager to apply strong problem-solving and software development skills to build reliable, practical systems.

Technical Skills

- **Programming Languages:** Python, C++, C, Java, Bash, SQL, JavaScript, C#
- **Software Development & Frameworks:** .NET, System-level programming, Real-time performance optimization, Networking protocols (TCP/IP, UDP), Algorithm design
- **Database Management:** MySQL, stored procedures, data integrity management
- **Frameworks & Libraries:** Matplotlib, NumPy, SymPy, React, .NET
- **Tools & Automation:** Cloud-based tools, Git, Linux/Bash automation scripting, Visual Studio, Visual Studio Code, Eclipse
- **Operating Systems:** Windows, Linux, Mac
- **Mathematics & Modelling:** Linear algebra, mathematical modelling
- **Teaching & Mentoring:** Python and SQL tutoring, curriculum development

Education

Bachelor of Science, Computer Science September 2020 - May 2026

Mount Royal University Calgary, AB

- **GPA:** 3.2 / 4.0
- **Achievements:** Mathematics Cognate, Baduk / GO AI Capstone Project, Linear Algebra Mathematics Research

Projects

Defender Remake on Atari ST

- Engineered a classic arcade game using C and assembly language. Optimized memory usage by implementing efficient system-level programming techniques on limited hardware

Companion Baduk

- Architected a desktop application integrating custom AI/ML move-evaluation algorithms, decision logic, and game analysis. Designed data pipelines with SQLite and Node.js to provide real-time performance insights and seamless interface rendering

Linux Shell Development

- Engineered a robust, custom Unix shell in C focusing on process management, resource control, and inter-process communication (IPC) to maximize execution reliability and maintain mission-critical stability under heavy load

Linear Algebra Research Project – PIMS Presentation & MRU Research Days

- Developed and presented a Python-based analysis of Anscombe's Quartet using Matplotlib, NumPy, and SymPy to demonstrate key statistical principles. Created visualizations and a detailed PowerPoint presentation to enhance audience understanding during the Pacific Institute for Mathematical Sciences event

Work Experience

Infinite Mind Pictures June 2025 - September 2025

Web Development Intern Calgary, AB

- Developed an internal wiki with Quartz 4 (React), enabling the team to share resources more efficiently and reducing duplicate documentation
- Enhanced user experience and visual design through custom TSX component edits and layout improvements
- Built an accessible resource portal for neurodivergent users using React, TypeScript

Launch Pad Learning May 2023 - January 2025

Programming Tutor Calgary, AB

- Facilitated hands-on Java coding sessions for 1-5 students (ages 10-30) using Eclipse IDE, and designed interactive activities that increased student engagement and completion rates
- Delivered the Python programming curriculum multiple times, enhancing students' problem-solving and coding skills
- Created custom JavaScript exercises that reinforced core programming concepts, leading students to demonstrate stronger understanding in subsequent assessments
- Coached students on coding best practices, resulting in improved comprehension and application of programming principles

Darwin's Moving & Deliveries July 2018 - August 2020

Mover & IT Support Calgary, AB

- Maintained network infrastructure and critical electrical/computer systems, ensuring uninterrupted uptime for operational logistics
- Provided IT support and rapid technical troubleshooting, reducing device downtime across hardware, software, and local networks